XRPL Validator — Daily Engineering Report · 2026-07-28

Period: 2026-07-28 00:00 to 24:00 US Eastern (2026-07-28 04:00 to 2026-07-29 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-28): 1.00000 (0 missed of 21,522). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

Earlier FFI shadow log (libxrpl applied against the validator's mirror; superseded by Step 1): 41 transaction-level differences recorded this day, across 1 ledger. These were not triaged into receipts.

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): honour the NFT offer opt-out on Owner and Destination** (`8b23677` , xrpl-validator) An account can opt out of receiving NFT offers with `lsfDisallowIncomingNFTokenOffer` (0x04000000), and rippled's shared `tokenOfferCreatePreclaim` (NFTokenHelpers.cpp:824) honours that for BOTH accounts an offer names: a `Destination` must exist (tecNO_DST), and neither it nor the token's `Owner` may carry the flag -- either is tecNO_PERMISSION. We checked neither.
- **fix(xrpl-ledger): finish CheckCreate's destination checks** (`edf4462` , xrpl-validator) Our CheckCreate preclaim stopped at "the destination exists". rippled's goes on to ask whether that destination will actually accept the check, and #105846674 F6CC9A594B17 is the case we were missing: a VRTY airdrop check to rnabZzjg, whose flags 0x120000 carry lsfRequireDestTag, sent with no DestinationTag. Mainnet claims the fee in one mutation; we created the Check and both directory entries in seven.
- **fix(xrpl-ledger): cap a direct IOU payment at what the destination can receive** (`cce3c8d` , xrpl-validator) rippled caps a payment's final issuer->dest step at `creditLimit(dst,issuer) - heldByDst` , floored at zero (DirectStepI::maxPaymentFlow, DirectStep.cpp:487), so a destination already at or over its own trust limit receives nothing and the strand is dry. `apply_path_payment` has carried that cap since 04a1586; that commit called the direct route the same rule "in principle" and left it alone for want of a failing …
- **feat(xrpl-ledger): token escrow, and give escrows their directory entries** (`5704d68` , xrpl-validator) #105823810 6AB38288 escrows 3750000 STSH. We rejected it temBAD_AMOUNT and applied nothing (0v6);

mainnet built it in 7 nodes. This is the escrow lead the handoff has carried for weeks -- it finally has a fixture.

- **fix(xrpl-ledger): an AMM deposit must fund its XRP side after reserve** (`11a9c6c` , xrpl-validator) #105893158 85C32164 (7v1). We had only the reserve guard from 385e129, which asks whether the depositor clears its reserve at all. rippled additionally checks that the XRP side is FUNDABLE, and measures that against the reserve the depositor will owe AFTERWARDS: `xrpLiquid` is taken with the owner count bumped by one when there is no LPToken line yet, because the deposit is about to open one (AMMDeposit.cpp:230-244, …
- **fix(differential_probe): put the RPC server in the cache key** (`52a567d` , xrpl-validator) The dxcache is one directory shared by every probe run on the box, and they do not all talk to the same node. The nightly scout hydrates from our own the reference node (`runner.sh diff <seq> http://[host]`); re-probing an older find has to use s2, because the reference node's ~17k-ledger window rolled past it long ago. The key was method+params only, so one server's answers were served for the other server's question.

Independence program

No independence record for UTC day 2026-07-28; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle Bonding & Slashing for Escrow Feeds** (oracles) Add sfOracleBond (drops) and sfChallengePeriod to the PriceOracle SLE, set via a new OracleBondSet transaction that locks a bond distinct from owner r

- **On-Ledger Finality Certificates** (consensus) Introduce a per-ledger singleton `ltFINALITY` object at keylet::finality(ledgerSeq) and a pseudo-transaction FinalityAttest that any validator master
- **Permissioned Iceberg Block Trading** (dex) Extend g31's tfIceberg OfferCreate flag with optional sfDomainID (referencing an existing PermissionedDomain) and sfMinimumFill SFields. When sfDomain

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-28 04:28	—	xrpl-validator	`8b23677`	fix(xrpl-ledger): honour the NFT offer opt-out on Owner and Destination
07-28 12:06	—	xrpl-validator	`edf4462`	fix(xrpl-ledger): finish CheckCreate's destination checks
07-28 12:38	—	xrpl-validator	`cce3c8d`	fix(xrpl-ledger): cap a direct IOU payment at what the destination can receive
07-28 13:08	—	xrpl-validator	`5704d68`	feat(xrpl-ledger): token escrow, and give escrows their directory entries
07-28 13:30	—	xrpl-validator	`11a9c6c`	fix(xrpl-ledger): an AMM deposit must fund its XRP side after reserve
07-28 14:13	—	xrpl-validator	`52a567d`	fix(differential_probe): put the RPC server in the cache key

Times in UTC. A commit appears on the day it was committed to main or merged into it.